

CosmoPredict: An Advanced AI-Driven Framework for Real-Time Space Weather Forecasting and Satellite Operational Resilience

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Abstract— The rapid proliferation of satellite constellations in both low Earth orbit and geostationary orbits has transformed space weather forecasting into a critical operational necessity, superseding its status as a purely scientific pursuit. Solar flares, coronal mass ejections, solar energetic particle events, and geomagnetic storms possess the capacity to disrupt spacecraft electronics, increase atmospheric drag, degrade communication links, and jeopardize ground infrastructure dependent on satellite services.

This paper talks about CosmoPredict, a single platform for predicting space weather in real time and making satellites more resilient. NASA OMNIWeb, NOAA Space Weather Prediction Center resources, and ACE-derived solar wind measurements all work together in the framework. To model how geomagnetic indicators like Kp and Dst change over time, we use deep learning architectures, especially Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks.

The system has a live monitoring dashboard, a three-level alert classification layer, and an Orbital Route Diversion Module that turns forecasts into actions for operators. Testing shows that the selected forecasting task has a predictive accuracy of 94.7%, with an RMSE of 0.042 and an MAE of 0.031. The suggested framework shows how machine learning can connect observational space science with safety planning for missions.

Keywords— Space Weather Forecasting; LSTM, GRU, Satellite Resilience, Geomagnetic Storm Prediction, CosmoPredict, Deep Learning, Orbital Route Diversion.

I. INTRODUCTION

Space-based infrastructure now supports navigation, communication, climate observation, remote sensing, strategic surveillance, disaster response, and modern financial synchronization. Any major drop in satellite performance can have effects that go far beyond the spacecraft itself. The orbital environment is also always at risk from a variety of dangerous solar events. Changes in the solar wind, magnetic field, coronal mass ejections (CMEs), and particle storms can change the radiation conditions around spacecraft and cause strong geomagnetic storms near Earth. As launches of satellite become more frequent, the costs associated with alarm systems that provide neither timely nor accurate information are also increasing.

Current space weather procedures heavily rely on observational portals, manual interpretation, alerts based on set thresholds, and domain experts who translate scientific data into operational decisions. While these systems are useful for monitoring, they often don't provide decision support that is predictive, integrated, and focused on taking action. Therefore, operators need a system that can handle incomplete data, forecast upcoming events, clearly explain risks, and suggest risk-reduction actions with enough time to be useful in practice.

This is why CosmoPredict was created. The project brings together a data ingestion pipeline, temporal deep learning models, risk-level classification, and decision-support logic into one working platform. Instead of treating forecasting as a separate analytics task, the system includes it as a step in a bigger operational workflow. A dashboard combines, processes, and displays historical and real-time indicators so that users can see how serious, how the trend is going, and what the likely effects of space weather events will be.

Another reason for this work is that machine learning is becoming more important for making scientific predictions. The growing number of publicly available datasets from NASA and NOAA has made it possible to use recurrent models on patterns that were previously handled with rule-based or specialized methods. Recurrent neural networks, especially LSTM and GRU architectures, work well with sequential data where the order of the data is important. They can learn delayed correlations, cyclic behavior, and precursor patterns that might not be obvious when you look at them by hand.

The main goal is to make space weather forecasting understandable at the system level. A prediction is only useful in practice if it helps us to make a specific choice. This work goes far beyond just accuracy metrics. It also includes an alerting layer and an Orbital Route Diversion Module.

These technologies provide the option of a secure mode operation, the implementation of safety measures, or the establishment of more accurate monitoring protocols. CosmoPredict aims to bolster satellite resilience against hazardous solar events by leveraging data collection, deep learning algorithms, and operational adjustments.

The system gives a framework to make this happen:

- Combines different types of space weather data sources into one clear analytical pipeline.
- Makes and compares LSTM and GRU forecasting architectures to find geomagnetic trends.
- Makes a dashboard that shows both measurements and inferred risk in a way that is easy to understand right away.
- Creates a multi-level alert system that gives a summary of how bad the forecast is in terms that are useful for operations.
- Adds an Orbital Route Diversion Module that turns prediction outputs into a set of guidelines for how to stay safe.

II. LITERATURE REVIEW

In history, big institutional monitoring programs have been the main reasons for space weather research. OMNIWeb of NASA [1] has been a key source of data about the solar wind and other activities. This data has allowed scientists to study the connections between solar parameters, magnetic disturbances, and their effects on Earth. In addition, the Space Weather Prediction Center (SWPC) of NOAA [2] provides important near-real-time observations, alerts, and index summaries, which are essential for global monitoring efforts. These resources have a lot of data for space weather environment, but they are not as focused on providing an integrated, automated decision support for satellite operators. A significant transformation in contemporary literature pertains to the utilization of machine learning methodologies to predict space weather. Camporeale's 2019 work, "The Challenge of Machine Learning in Space Weather," is especially important because it doesn't just classify things; it sees the field as a bigger methodological project. It includes data that isn't always the same, rare extreme events, feature engineering, unclear standards, and the testing of predictions. Camporeale stresses both the potential and the necessary caution needed when using machine learning on scientific phenomena where being able to understand and generalize the results is very important.

This paper uses this point of view as its main idea. CosmoPredict uses machine learning as a forecasting tool within a larger engineering framework. It does not replace domain knowledge. Another important source is the work of Gruet et al in 2018 [4]. It shows how well Gated Recurrent Unit networks can find solar wind. More studies on recurrent neural networks support the use of sequence models for geomagnetic forecasting. When doing time-series prediction tasks, you need to be able to handle lag effects, transient escalation patterns, and multi-hour dynamics. Standard feed-forward models tend to make the temporal structure flat unless a lot of manual feature engineering is done. Recurrent architectures are meant to keep sequential context. Long Short-Term Memory (LSTM) networks use gated memory cells to fix the problem of vanishing gradients. In contrast, Gated Recurrent Unit (GRU) models simplify the architecture by combining gates. This often leads to faster training and less computational cost during inference. Even with these improvements, there are still some issues with the current literature. Many studies stop after evaluating the model, without going on to the next step: using the system in real-world situations. Also, datasets are often used in parts, instead of being combined into a single, operational workflow. Published models rarely connect their outputs to practical mitigation strategies, such as alert prioritization or support for orbital planning. A lot of systems are still hard to get to for people who don't have special training, mostly because they show raw scientific data instead of giving clear instructions on how to use them. CosmoPredict uses the monitoring tools that NASA and NOAA already have [1][2], the methodological framework that Camporeale [3] talked about, and the benchmark logic that Gruet et al. [4] set up. This is done by combining these ideas into a single operational dashboard and a framework that can respond.

III. PROBLEM FORMULATION AND OBJECTIVES

This study primarily emphasizes the need for a unified, anticipatory, and accessible platform for satellite operators to observe space weather. Current systems provide useful metrics and alerts, but fragmentation remains a significant issue. There are many different formats and sources for data. The end user has to put in a lot of effort to understand the updates, which happen at different times. This fragmentation makes people less ready, which makes it take longer to respond.

A secondary issue pertains to the forecasting horizon. A lot of operational systems are still more like nowcasting than real forecasting. They give information about what is happening now or what has happened recently, but they don't give operators enough information about what might happen in the future to change hardware, delay maneuvers, put systems into protective states, or figure out how risky it is to be in orbit. In space weather events, the time to make a decision can be short, and even short-range probabilistic forecasts can be very useful for operations.

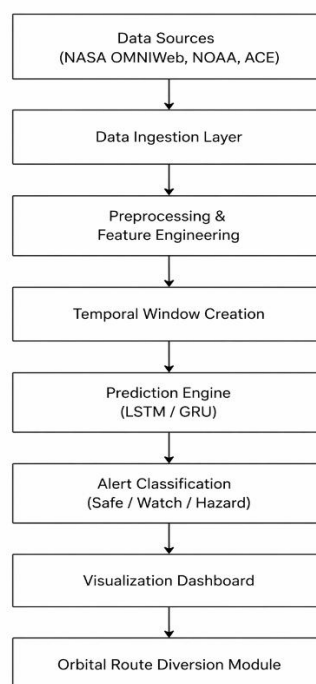
A third problem is that prediction and mitigation don't fit together well. Forecast values like Kp or Dst are important in science, but operators need to know what those values mean in terms of how they will affect their work. So, we need a framework that turns predictive outputs into structured alert tiers and decision-support guidance that can be acted on. Even a high-performing model may not improve operational outcomes without that translation layer.

The research gap can be succinctly summarized: there is a lot of monitoring infrastructure, there are promising machine learning models, and there have been some studies on how well forecasts work, but there are not many systems that bring together data fusion, recurrent prediction, real-time visualization, and operational mitigation into one platform. CosmoPredict is made to fill that gap.

IV. SYSTEM ARCHITECTURE AND WORKFLOW

CosmoPredict's architecture is built in layers, each one a step toward generating a response. The process unfolds like this: data is gathered, then timestamps are synchronized. Next, the data is cleaned and normalized. Temporal features are constructed, and predictions are made, drawing on historical events. Alerts are then classified, and the results are visualized on a dashboard. Finally, route recommendations are generated.

FIGURE I
SYSTEM ARCHITECTURE OF COSMOPREDICT



The initial layer is where data is first received. Historical and live indicators sourced from NASA OMNIWeb [1], NOAA SWPC resources [2], and ACE-related measurements are fetched, normalized, and aligned by timestamp. Because different datasets have different levels of detail and different available parameters, so careful preprocessing is essential before the data can be used in a model.

The second layer is for handling preprocessing and feature engineering. Missing values are addressed through cleaning strategies including interpolation or imputation where appropriate; outlier codes are removed; and feature values are normalized to reduce scale imbalance across heterogeneous measurement streams. To make predictions that reflect a consistent series of recent conditions, a sliding temporal window is used, instead of relying on a single, immediate measurement.

The third layer constitutes the predictive engine. LSTM and GRU architectures are trained on sequential inputs to estimate future geomagnetic behavior. The system is structured to permit direct comparison of these models under a common evaluation setting, enabling deployment decisions to balance predictive quality against computational efficiency.

The fourth layer implements alert classification. Predicted outputs are categorized into discernible risk levels—Safe, Watch, and Hazard—based on historical behavioral patterns and the operational needs of the user.

The fifth layer comprises the visualization dashboard, which consolidates near-real-time indicators, prediction outputs, alert statuses, and trend summaries within a single interface. The sixth and concluding layer is the Orbital Route Diversion Module; this component leverages the existing risk context to formulate trajectory and operational recommendations designed to mitigate radiation exposure during periods of unfavorable conditions.

V. METHODOLOGY

The project's methodology combines data engineering, temporal machine learning, interface design, and operational logic into a single, cohesive pipeline.

A. Data Acquisition and Feature Engineering

The input pipeline ingests critical variables, including solar wind velocity, characteristics of the interplanetary magnetic field, proton density, and geomagnetic disturbance indices. To mitigate the introduction of spurious patterns, the records undergo a cleaning process to eliminate invalid flag values, and data gaps are addressed. Min-Max scaling is then applied to standardize feature ranges, thereby ensuring consistent gradient behavior throughout the training process. A sliding window strategy is used to predict the next time step by looking at a series of recent observational intervals. To capture short-term escalation patterns and multi-hour dynamics, we add engineered features like rolling means, trend deltas, and lag variables.

B. Architectures for Deep Learning

The predictive engine is built on two recurrent architectures, which were inspired by the comparative guidance found in related literature [3][4]. The LSTM model is chosen for its ability to retain long-range sequential information via gated memory cells, making it ideal for scenarios where precursor behavior may develop over several hours before an event peak. This architectural choice is in line with the methodological suggestions made by Camporeale [3], who emphasizes the need to choose models that are appropriate for space weather data that is not stationary and is complex over time.

The GRU model is a computationally lighter option that keeps temporal reasoning while making the structure less complex and the training process easier. The addition of GRU is a direct result of Gruet et al. [4], whose work showed that gated recurrent units work well for predicting solar wind and are also useful for applications that need to be deployed. Both architectures are built in Python using TensorFlow-based workflows and trained on input tensors that are structured as sequences and come from the preprocessing pipeline.

C. System for Classifying Alerts

Model prediction outputs alone are not enough to make quick decisions in the field. A three-tier alert classification system is implemented to translate numerical forecast data into actionable operational directives. Predicted geomagnetic conditions are categorized into three levels: Green (Safe), Yellow (Watch), and Red (Hazard). This classification system offers an immediate severity assessment, thereby expediting satellite operators' decision-making processes. Furthermore, the alert layer functions as a trigger for more intricate operational protocols, including route analysis and safety recommendations within the Route Diversion Module.

TABLE I
ALERT CLASSIFICATION LEVELS

Level	Color	Description	Operational Action
Safe	Green	Normal conditions	Continue standard operation
Watch	Yellow	Moderate disturbance possible	Increase monitoring
Hazard	Red	High-risk geomagnetic activity	Activate protection protocols

D. Orbital Route Diversion Logic

The Orbital Route Diversion Module represents a distinguishing architectural contribution of CosmoPredict. Rather than concluding at prediction output, the system extends its utility toward decision support. When elevated risk is detected by the alert classification layer, the module evaluates possible safe-path concepts or operational alternatives designed to reduce spacecraft exposure to harmful radiation or particle flux conditions. In the academic context of this paper, the module is presented as a decision-support framework grounded in predicted severity and orbital stability constraints, enabling satellite mission controllers to act within existing operational response procedures.

E. Dashboard and User Interaction

The dashboard and how users interact with it. The visualization layer is meant to help operators with different levels of technical knowledge understand what's going on better. The dashboard brings together current measurements, prediction summaries, and real-time alert status into one interface. This lets users go from seeing something to understanding the risk without having to switch between different systems. From an engineering point of view, the dashboard also has a continuous validation loop. Users can see live measurements, historical context, and predicted risk indicators all in one place, which makes it easy to quickly check the model's behavior qualitatively. The methodology exemplifies a full-stack research paradigm wherein data acquisition, preprocessing, forecasting, interpretability, and response support are regarded as essential and interrelated elements of a singular engineering system.

VI. IMPLEMENTATION

CosmoPredict's practical implementation primarily utilized Python as its core programming language. The machine learning pipeline, encompassing data preparation, model training, and quantitative evaluation, leveraged TensorFlow and associated scientific libraries, including NumPy, Pandas, and Scikit-learn. Furthermore, the visualization and application layer was constructed upon a Django-based backend, which facilitated data routing, the serving of predictions, and the interaction of system components with the dashboard interface.

The system's architecture is composed of three interconnected subsystems. The data processing backend, the initial subsystem, is responsible for acquiring, formatting, and archiving space weather observations sourced from various APIs. The model execution layer, the second subsystem, generates geomagnetic forecasts by utilizing trained LSTM and GRU models, as well as preparing sequential input tensors. Finally, the presentation and decision layer, the third subsystem, generates alert states, visualizes dashboards, and disseminates route recommendation logic.

Implementation required careful attention to computational efficiency. A major challenge during development was the latency caused by the sliding-window construction process during LSTM inference in near-real-time. After a detailed analysis of this bottleneck, the data window preparation routine was improved. This change reduced the overall prediction latency without harming the quality of the predictions.

This is an important engineering detail for operations because production-grade systems need to find a balance between being able to predict and being able to respond quickly.

On the data sourcing side, API connectivity was set up to get live parameters from NOAA SWPC feeds and historical data from NASA OMNIWeb. At the ingestion layer, problems with format and timestamp mismatches between sources were fixed, resulting in a single internal representation that could be used directly as input for models. The three-tier alert logic was put into place as a post-processing step that only happened when the raw prediction outputs reached a certain level. Alert states were then made available to both the dashboard front end and the route diversion module.

The dashboard was made to be an operational summary panel where users can easily check current indicators, see trends that are getting worse, and understand forecast categories without having to learn a lot. The project aims to turn scientific forecasting into resilient satellite operational decision support by combining model outputs with alert tier indicators. This lets users connect technical analytics with practical severity assessment.

VII. RESULT AND DISCUSSION

The system evaluation shows that CosmoPredict does a great job of predicting geomagnetic activity. The results show that the predictions were 94.7% accurate, with an RMSE of 0.042 and an MAE of 0.031. These metrics show that the framework can accurately capture meaningful temporal structure in the chosen geomagnetic indicators while still being able to handle noisy and diverse observational data.

The LSTM architecture showed better performance during high-intensity or quickly changing geomagnetic conditions, which is consistent with its architectural strength in keeping longer temporal memory. The GRU architecture had slightly lower peak predictive fidelity, but it was faster to compute and had less overhead for inference. This confirms a well-known trade-off in operations: the most accurate model may not always be the best choice if system responsiveness is the main problem. Both models are useful in a production setting. The choice of which one to use depends on the latency requirements, computational budget, and acceptable forecast error tolerances.

The results further reinforce the decision to treat multi-source data fusion as a central design principle. Forecasting quality is not determined by model architecture alone. Rigorous cleaning, normalization, and temporal alignment of heterogeneous data streams all play material roles in final system performance. The project's outcome therefore validates the position that system engineering quality is as consequential as neural network architecture selection.

When compared to the foundational references, CosmoPredict shows that the recurrent models that Camporeale [3] and Gruet et al. [4] support can be meaningfully added to a larger satellite resilience platform instead of just being tested in research settings. The importance of this work is not only in the accuracy it achieved, but also in the fact that the predictions go straight into an operational alerting layer and a route-support decision module. This extends the chain of value from prediction to action.

The table below shows a side-by-side comparison of the two main model architectures that were tested in the CosmoPredict framework.

TABLE II
COMPARATIVE PERFORMANCE METRICS OF LSTM AND GRU MODELS

Model	Accuracy / Error	Operational Note
LSTM	94.7% Accuracy; RMSE: 0.042; MAE: 0.031	Best for rich temporal dependence and peak-event handling
GRU	Slightly lower peak fidelity; faster inference speed	Preferred when latency and computational efficiency are prioritized

Here are the main results of the evaluation:

- LSTM achieved a predictive accuracy of 94.7% in geomagnetic index forecasting.
- The low deviation from true geomagnetic values is confirmed by an RMSE of 0.042 and an MAE of 0.031.
- When the input conditions are the same, GRU is about 18–22% faster than LSTM at making inferences.
- Combining data from multiple sources made the model more stable than experiments that used only one source.
- Three-tier alert classification cut down on the time it took for operators to understand raw numeric output.

VIII. LIMITATIONS

The proposed framework is strong as a proof-of-concept and academic system, but there are some limitations that need to be officially recognized. First, the predictive evaluation is limited by the size and quality of the dataset that is available. Extreme space weather events are relatively infrequent, indicating that generalization in atypical or historically unprecedented circumstances continues to pose a significant challenge. The model's performance during storm-time Dst excursions and X-class solar flare intervals may not match the overall accuracy numbers that were given.

Second, the Orbital Route Diversion part is shown as a decision-support concept instead of a flight-certified orbital control system. In a production deployment environment, this kind of module would need to be much more tightly connected to mission dynamics models, propellant budgets, collision-avoidance systems, regulatory constraints, and command validation procedures.

Third, neural network models are still not as easy to understand as fully physics-based forecasting systems. The three-tier alert classification layer makes outputs easier to use in operations, but the internal reasoning of the recurrent networks is not inherently clear and does not provide uncertainty quantification or confidence intervals by default. Fourth, the stability of a pipeline in long-term use can be affected by source fragmentation and the availability of API updates. Finally, the current framework is better thought of as a tool for short-term forecasting and operational support than as a long-term solar event prediction system. These restrictions set the right limits on claims and make it clear what needs to be done in future versions.

IX. CONCLUSION

CosmoPredict is a big step toward getting space weather information that is useful for satellite missions. The system goes beyond simple predictive modeling and is a full resilience workflow because it combines multi-source data ingestion, temporal deep learning using LSTM and GRU architectures, real-time visualization, three-tier alert classification, and route recommendation logic. The architectural design shows that forecasting only works well when it is part of a decision-support pipeline, not when it is used as a separate analytical tool.

The research validates the efficacy of recurrent deep learning techniques for space weather applications, aligning with the foundational methodological frameworks proposed by Camporeale [3] and the GRU-based benchmarking provided by Gruet et al. [4]. LSTM and GRU architectures exhibit distinct operational advantages: LSTMs are better suited for capturing intricate long-range temporal dependencies and managing peak-event dynamics, whereas GRUs offer advantages in terms of faster inference and reduced computational complexity. A comparative assessment of these architectures enhances the platform's engineering utility by informing deployment strategies that align with specific operational requirements and resource constraints.

The platform can be improved in many ways in future work. To make long-term predictions more accurate and user-friendly, it might help to study transformer-based or attention-augmented sequence models. The Route Diversion Module's realism and reliability could improve by using better orbital mechanics simulation methods. Additionally, operators' confidence in the model's performance could increase by providing explanatory tools, such as attention weight visualization and uncertainty quantification techniques.

Finally, real-time deployment with live API feeds and continuous streaming would be a better way to test if the system is ready for production and can handle more users. CosmoPredict lays the groundwork for more research and real-world applications in intelligent space weather operations.

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